

REVIEW 1

GUIDE:DR.VANACHARLA PUJITHA

GROUP MEMBERS INFO:

- 1)23BCE7268-PULAGAM MAHESWARA REDDY
- 2)23BCE20147-KARUMBURU NITIN
- 3)23BCE20245-OJILI SUBODH
- 4)23BCE20265-KOPPALI RANJITH
- 5)23BCE20329-THUPAKULA REVANTH REDDY
- 6)23BME7083-SHAIK ABDUL RAHEEM

An aerial photograph of a winding asphalt road that curves through a dense, lush green forest. The road is light gray and contrasts with the dark green foliage. A semi-transparent green rectangular overlay covers the left side of the image, serving as a background for the text. The overall composition is clean and modern, with a focus on nature and infrastructure.

CONTRETEMPS NULLIFICATION

Problem Statement:

- **Safety Risk:** Blind spots increase accident risks, especially in hilly and wooded areas.
- **Limited Visibility:** Curving roads and obstructions like hills and trees hinder clear sight.
- **High Collision Risk:** Lane changes and turns in restricted areas often lead to accidents.



Objective:

- **Enhance Safety:** Reduce collision risks by providing timely alerts.
- **Cost-Effective:** Develop an affordable and efficient detection system.
- **Wide Usability:** Ensure functionality in both urban and rural environments.

Target Outcome:

- **Increased Road Safety:** For drivers in challenging terrains.
- **Accessible Technology:** Easy integration into vehicles for broader adoption.

Advantages:

- **Enhanced Driver Awareness:** Provides real-time alerts for safer lane changes and turns.
- **Reduces Accidents:** Lowers the risk of collisions in areas with limited visibility.
- **Cost-Effective Solution:** Affordable technology that can be widely implemented.
- **Versatile Application:** Effective in various terrains, including hilly and urban areas.
- **Easy Integration:** Compatible with existing vehicle systems, facilitating broader adoption.
- **Promotes Safer Roads:** Contributes to overall road safety, benefiting both drivers and pedestrians.

Driver warning system in hill road bends for safer journey



Required Components:

- ESP-32-
- Ultrasonic Sensor
- Active Buzzer
- Red LED
- 3.7v Battery
- Power Booster Module
- Switch



Approximate Budget Required:

- ESP-32 :2000
- Ultrasonic Sensor:600
- Active Buzzer:150
- Red LED:100
- 3.7v Battery:150
- Power Booster Module:150
- Switch:50
- TOTAL BUDGET:35000



THANK YOU

